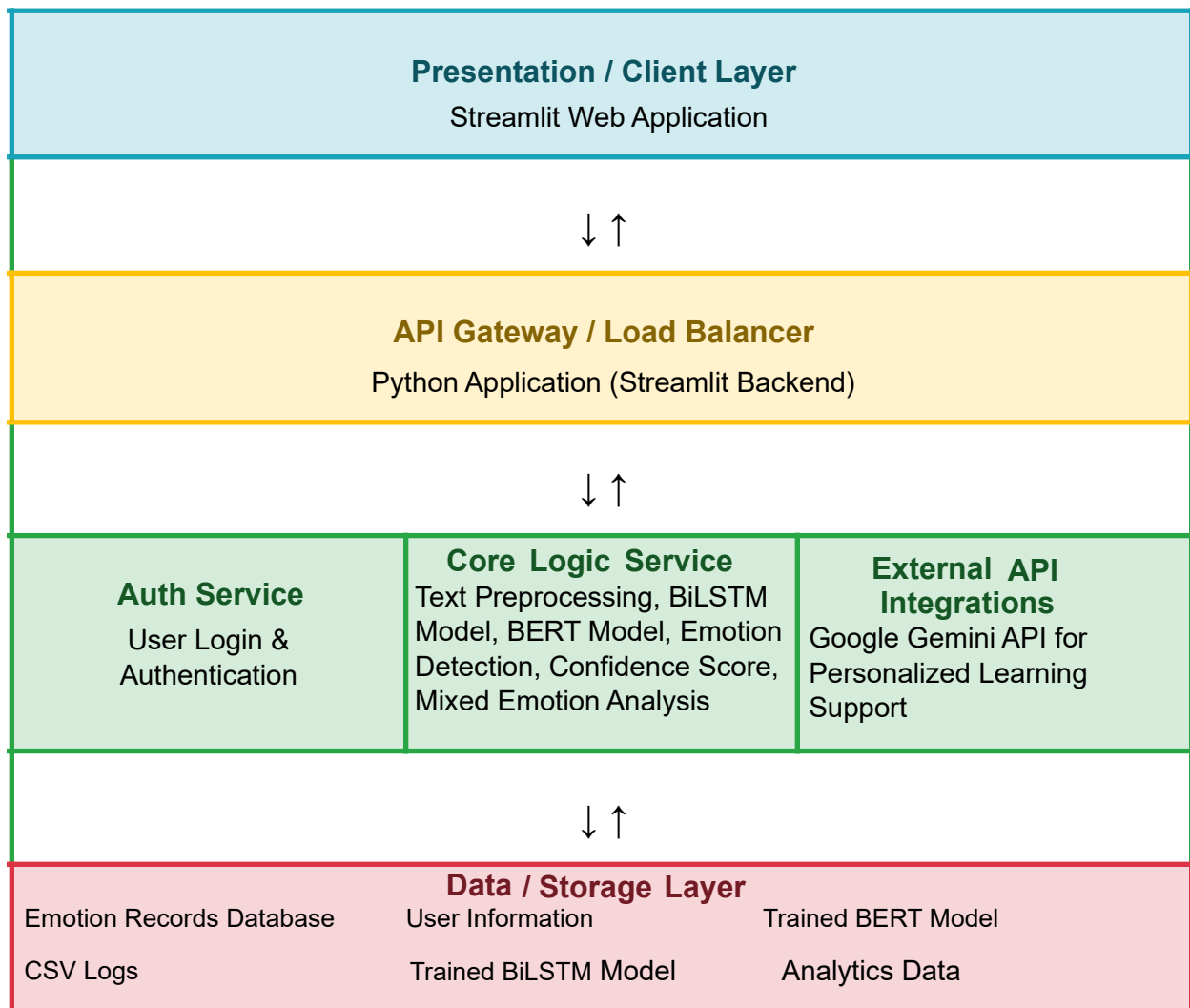


Solution Architecture

Date	28-06-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where students and educators can submit learning problems, view detected emotions, AI-generated guidance, and analytics.	<i>Streamlit, HTML, CSS</i>
API Gateway / Application Layer	<i>Handles incoming requests, validates user input, manages application workflow, and connects the frontend with AI services.</i>	<i>Python, Streamlit</i>
Authentication Service	<i>Manages user registration, login, authentication, and session handling to ensure secure access to the application.</i>	<i>Python, SQLite / Local Authentication</i>
Core Logic Service	<i>Performs text preprocessing, emotion detection using BiLSTM and BERT, calculates confidence scores, and supports mixed emotion prediction.</i>	<i>TensorFlow, Keras, Hugging Face Transformers, NLTK</i>
External API Integration	<i>Connects with Google Gemini AI to generate personalized learning guidance based on the detected emotion.</i>	<i>Google Gemini API</i>
Data / Storage Layer	<i>Stores user details, emotion records, AI responses, timestamps, CSV logs, trained models, and analytics data for future reference.</i>	<i>CSV Files, Local Storage</i>